

Steering Optical Signals In The Visible Light Spectrum Using Beamforming

Background

Many modern systems in many different fields involve steering electromagnetic wavefronts at various different wavelengths. At smaller wavelengths, the techniques for steering these electromagnetic waves fall under the category of optics, and the steering is typically carried out with specialized tools called lenses, which are specialized materials in specific shapes that affect the velocity of the wave moving through in such a way as to change the direction of the wave propagation. In many circumstances, lenses can present their own engineering challenges, such as heating under high energy, and can take a lot of space in space-constrained applications.

In larger wavelength electromagnetic wave applications, such as radar, and in acoustic applications, there is another common technique for steering the propagation of waves, called beamforming. Beamforming is a signal processing technique that steers the wavefront by slightly altering the phase of a given signal broadcast from multiple antennas in a single array, causing constructive interference in the desired direction, and destructive interference in every other direction [1].

While beamforming is an established signal processing technique, and used widely in many fields, in visible light applications it is mainly used for Visible Light Communication [2]. This application is useful, but low power and limited in scope. More general applications, like replacing optical systems, have had not had very much exploration.

This project aims to apply beamforming techniques to visible light, in order to steer the visible light in a way similar to optical lenses, to attempt to create an optical system that does not require physical lenses.

Question

Can beamforming techniques be used to steer visible light signals, and can this be used as a suitable alternative to physical optics?

Hypothesis

It is possible to steer visible light signals using beamforming techniques, and this can be used as a suitable alternative to physical optics in some applications.

Methods

The first part of this experiment will involve developing computer simulations of the wave propagation of visible light. This will lead into simulating a physical source array, and these simulations will be used to develop a model for a physical array. Twenty hours across the span of two weeks will be allotted for the development and validation of these simulation models.

Next, the physical array will be chosen and constructed. This will likely consist of some sort of LED display, but the simulations will provide more exact array implementation details. The array construction will also include a light sensor array[3], to quantifiably validate the beamforming results. The construction and testing of this setup will take an additional twenty hours across two weeks.

The third step of this experiment will involve writing a program to turn the simulated beamforming steering math into some instructions the array can actually execute. This will involve writing code to individually control subsections of the array, and to pulse those sections with the frequency and phase shifts necessary for beamforming to take place. Presumably this will be the most complicated part of the project, and forty hours will be allotted to this across the span of four weeks.

The fourth step of this experiment involves quantifiably validating the experimental results, and will involve writing a program to read from the aforementioned light sensor and log the recieved light data at different locations relative to the array. The point of this is to show that beamforming is actually occurring, and that will be shown by an increase in brightness at the intended location. This will take an additional twenty hours over two weeks.

Finally, the results will be analyzed, and a final report and presentation will be written to sum-

marize the findings of the experiment. Another twenty hours across two weeks will be allotted for this.

Significance

The aim of this experiment is to determine if beamforming techniques could be used to supplant traditional optical systems. If this is the case, many optical applications can become very simplified and much more efficient. This technique could be useful in heads up displays, virtual reality, telescopes, and even Extreme UltraViolet (EUV) semiconductor manufacturing. The benefits of this could be far reaching, reducing cost and leading to the development of new, simpler, and more efficient technologies.

References

- [1] Marc Lichtman. PySDR: A Guide to SDR and DSP using Python. <https://pysdr.org/>, 2023.
- [2] Satvik Gupta, Debangana Roy, Siddharth Bose, Vipul Dixit, and Atul Kumar. Illuminating the future: A comprehensive review of visible light communication applications. *Optics & Laser Technology*, 177:111182, 2024.
- [3] Analog Devices. <https://www.analog.com/en/resources/glossary/light-sensor.html>.